EXPERIMENT - 9

NAME: PARAMESWARI P ROLL NO: 240701378

Design input forms that validate data (e.g., email, phone number) and display error messages.

AIM:

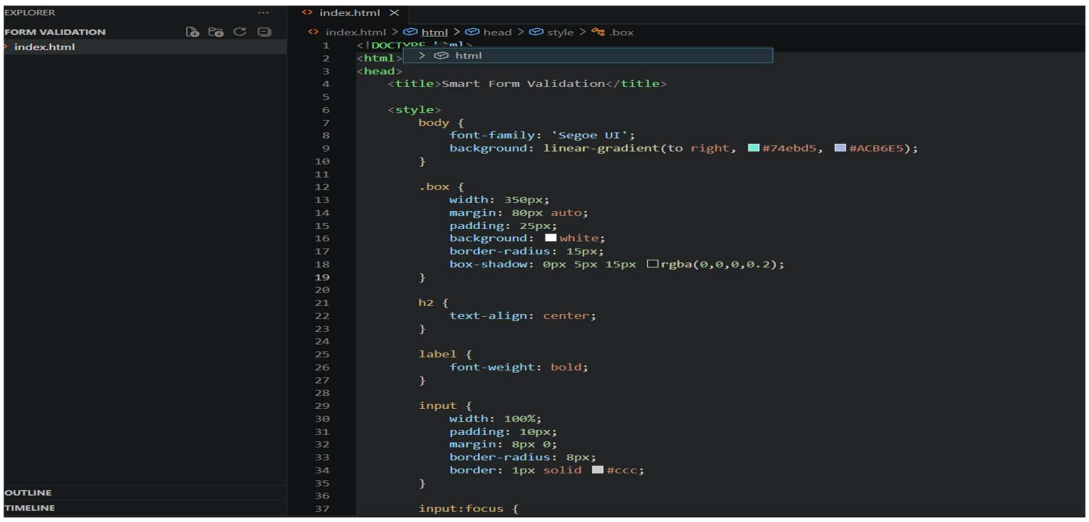
The aim of this experiment is to design a user-friendly registration form that validates input fields such as email and phone number, and displays appropriate error or success messages using HTML, CSS, and JavaScript.

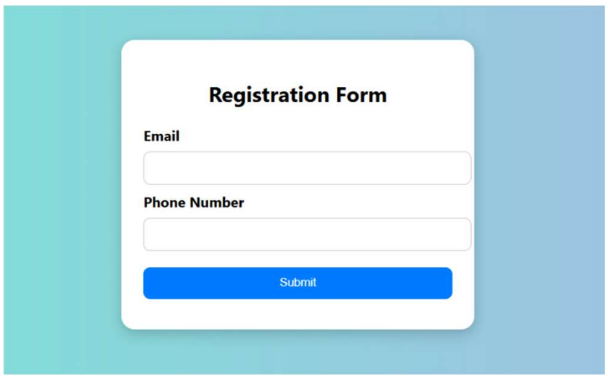
Step 1: Creating the HTML Form

Create a basic registration form with input fields for email and phone number along with a submit button. Also include space to display error and success messages.

Output for Step 1:

(Simple form with Email, Phone Number, and Submit button displayed)





Step 2: Styling the Form using CSS

Apply CSS styling to improve the appearance of the form. Add background gradient, spacing, rounded corners, and hover effects for better user experience.

Output for Step 2:

(Styled form with attractive layout, colors, and centered design)

Step 3: Adding JavaScript Validation

```javascript
function validatePhone() {
    let phone = document.getElementById("phone").value;

    if (!/^[0-9]{10}$/.test(phone)) {
        document.getElementById("phoneError").innerText = "Phone must be 10 digits";
        return false;
    } else {
        document.getElementById("phoneError").innerText = "";
        return true;
    }
}

function validateForm() {
    let isEmailValid = validateEmail();
    let isPhoneValid = validatePhone();

    if (isEmailValid && isPhoneValid) {
        document.getElementById("successMsg").innerText = "Form Submitted Successfully!";
        return false; // stop actual submit (for demo)
    }
    return false;
}

</script>

</body>
</html>
```

```html
<div class="box">
    <h2>Registration Form</h2>

    <form onsubmit="return validateForm()">

        <label>Email</label>
        <input type="text" id="email" onkeyup="validateEmail()">
        <div id="emailError" class="error"></div>

        <label>Phone Number</label>
        <input type="text" id="phone" onkeyup="validatePhone()">
        <div id="phoneError" class="error"></div>

        <button type="submit">Submit</button>

        <div id="successMsg" class="success"></div>

    </form>
</div>

<script>

function validateEmail() {
    let email = document.getElementById("email").value;
    let pattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

    if (!pattern.test(email)) {
        document.getElementById("emailError").innerText = "Invalid Email Format";
        return false;
    } else {
        document.getElementById("emailError").innerText = "";
        return true;
    }
}
```

Use JavaScript to validate the input fields:

Email is checked using a pattern (regular expression)

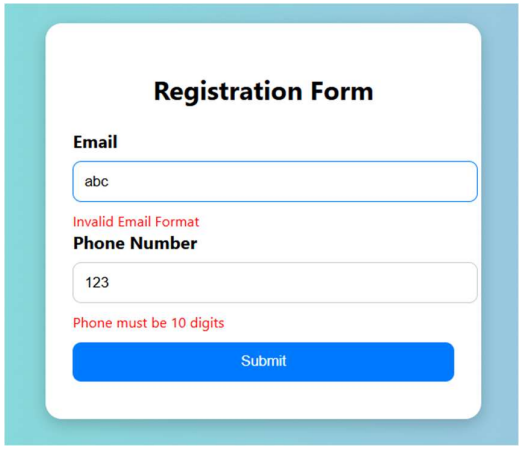
Phone number must contain exactly 10 digits

Display error messages if inputs are invalid

Show success message if all inputs are valid

Output for Step 3:

Case 1: Invalid Email and Invalid Phone Number



Email: abc

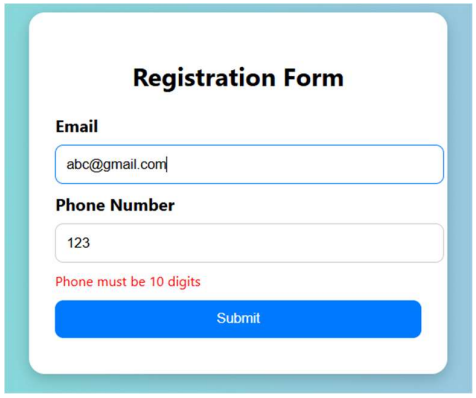
Phone: 123

Output:

Displays "Invalid Email Format"

Displays "Phone must be 10 digits"

Case 2: Valid Email and Invalid Phone Number



Email: abc@gmail.com

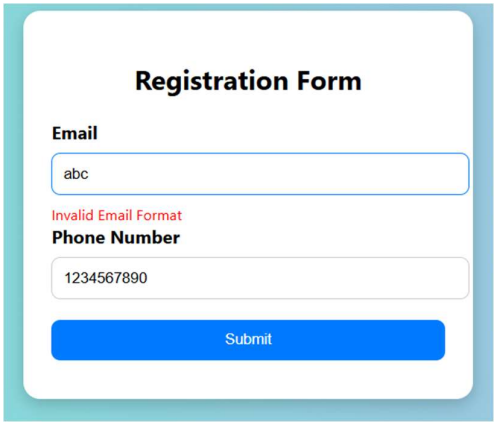
Phone: 123

Output:

Email accepted

Displays "Phone must be 10 digits"

Case 3: Invalid Email and Valid Phone Number



Email: abc

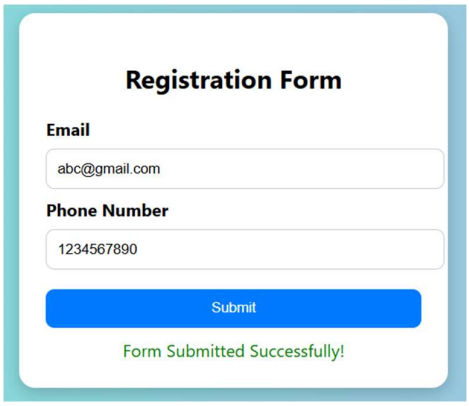
Phone: 1234567890

Output:

Displays "Invalid Email Format"

Phone accepted

Case 4: Valid Email and Valid Phone Number



Email: abc@gmail.com

Phone: 1234567890

Output:

Displays "Form Submitted Successfully!"

RESULT:

Thus, the input form was successfully designed with validation for email and phone number, and appropriate error and success messages were displayed using HTML, CSS, and JavaScript.